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River network architecture, genetic effective size and distributional patterns predict differences in genetic structure across species in a dryland stream fish community

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Abstract

Dendritic ecological network (DEN) architecture can be a strong predictor of spatial genetic patterns in theoretical and simulation studies. Yet, interspecific differences in dispersal capabilities and distribution within the network may equally affect species' genetic structuring. We characterized patterns of genetic variation from up to ten microsatellite loci for nine numerically dominant members of the upper Gila River fish community, New Mexico, USA. Using comparative landscape genetics, we evaluated the role of network architecture for structuring populations within species (pairwise F_{ST}) while explicitly accounting for intraspecific demographic influences on effective population size (N_e). Five species exhibited patterns of connectivity and/or genetic diversity gradients that were predicted by network structure. These species were generally considered to be small-bodied or habitat specialists. Spatial variation of N_e was a strong predictor of pairwise F_{ST} for two species, suggesting patterns of connectivity may also be influenced by genetic drift independent of network properties. Finally, two study species exhibited genetic patterns that were unexplained by network properties and appeared to be related to nonequilibrium processes. Properties of DENs shape community-wide genetic structure but effects are modified by intrinsic traits and nonequilibrium processes. Further theoretical development of the DEN framework should account for such cases.

Keywords: dendritic ecological networks, effective population size, extrinsic and intrinsic factors, gene flow, Gila River, landscape genetics

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Introduction

Dendritic ecological networks (DENs) are bifurcating habitat networks that constrain dispersal of organisms such as those found in river drainages (Grant *et al.* 2007), and have contributed to our understanding of relationships between landscape structure and ecological and evolutionary dynamics (Fagan 2002; Brown & Swan 2010; Grant 2011; Dias *et al.* 2013). The hierarchical distribution of habitat patches linked by linear corridors creates complex spatial arrangements that differ from lattice network geometry in how species disperse from one patch to another (Fagan 2002; Labonne *et al.* 2008; Grant 2011). The abiotic backdrop of DENs can be a primary driver of community composition and species distributions (Hughes *et al.* 2009; Rodriguez-Iturbe *et al.* 2009) as well as patterns of diversity across

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biological levels from communities (Carrara *et al.* 2012) to population genetic diversity (Hughes *et al.* 2009; Finn *et al.* 2011).

Theoretical and simulation studies on network geometry have reported that dendricity influences patterns of genetic variation, and in some cases, can increase genetic divergence relative to lattice or linear geometries (Thomaz *et al.* 2016). Furthermore, increased isolation of populations in distal branches of the network (i.e. headwater streams) and dispersal asymmetries can result in species exhibiting genetic diversity gradients (Morrissey & de Kerckhove 2009; Paz-Vinas & Blanchet 2015). However, linking genetic patterns to dendritic structure in empirical studies is not straightforward because species' dispersal abilities may overcome the constraints of the network (Chaput-Bardy *et al.* 2009). This has been observed in species with overland dispersal such as aquatic macroinvertebrates with flying adult stages (e.g. Alp *et al.* 2012; Phillipsen *et al.* 2015), or amphibians with low water dependencies (Mims *et al.* 2015). Alternatively, longitudinal habitat changes within the network can effectively isolate populations of headwater specialists (Finn *et al.* 2006, 2007) resulting in genetic patterns unrelated to the network. Nonequilibrium metapopulation dynamics and colonization history may also exert stronger influence over species genetic patterns than network architecture (in fish Salisbury *et al.* 2016; salamanders Cosentino *et al.* 2012). Lastly, demographic properties such as differences among local population effective sizes can lead to spatial genetic patterns induced by genetic drift (Weckworth *et al.* 2013). Thus, further evaluation of species with different life history traits within a shared dendritic network might help refine predictive models of how genetic diversity superimposes on network structure.

In this paper, we used a combination of landscape genetics and trait-based comparisons from nine co-occurring fish species, seven native and two introduced, to evaluate the influence of riverscape network features and life history on gene flow and genetic structuring in a stream fish community. Specifically, our objectives were to (i) quantify spatial genetic variation for multiple species of a stream fish community, (ii) characterize the role of landscape and genetic drift on population structure of each species using landscape genetics techniques and (iii) combine results of genetic analyses with ecological understanding of life history to evaluate the relative roles of intrinsic (e.g. species-specific traits) and extrinsic (e.g. river network features) factors in shaping genetic diversity.

The Gila River fish community is dominated by representatives of three orders (Cypriniformes, Siluriformes and Perciformes) and comprised of seven native fish species, *Agosia chrysogaster* (longfin dace), *Gila nigra* (headwater chub), *Meda fulgida* (spikedace), *Rhinichthys osculus* (speckled dace), *Tiaroga cobitis* (loach minnow), *Catostomus insignis* (Sonora sucker) and *Pantosteus clarkii* (desert sucker), and two species introduced within the last 100 years, *Ameiurus natalis* (yellow bullhead catfish) and *Micropterus dolomieu* (smallmouth bass). In general, relative abundances of species reflect trophic variation ranging from most abundant algivores/detritivores (*C. insignis* and *P. clarkii*) and insectivores (*A. chrysogaster* and *M. fulgida*) to less abundant apex predators (*A. natalis* and *M. dolomieu*) (Pilger *et al.* 2010). Body size, life history strategies (Olden *et al.* 2006), habitat preference and specificity (Stefferud *et al.* 2011; Whitney *et al.* 2014) and distribution within the network (Propst *et al.* 2008) all vary substantially across species (Table 1; Fig. 1). We hypothesized that network attributes such as stream distance between sites, change in elevation within the network and number of confluences would be strongest predictors of genetic structure for species with limited dispersal within the network. Because nearly 50% of dispersal propensity of lotic fishes is explained by body size (Radinger & Wolter 2014), we predicted that large-bodied species (i.e. *C. insignis*, *P. clarkii*, *A. natalis* and *M. dolomieu*) would exhibit less spatial structure due to high gene flow, whereas small-bodied species were predicted to be more spatially structured. We also predicted that generalist species (i.e. *A. chrysogaster*, *C. insignis* and *A. natalis*) would exhibit less genetic differentiation than habitat specialists because generalists are less likely to be constrained by habitat patches within the network. However, patterns of connectivity may also depend on species' distribution within the network (Hughes *et al.* 2009). For example, we predict that widely distributed and abundant species should have less spatial structuring among populations than species with patchy, disjunct distributions or those restricted to headwaters. Lastly, because genetic diversity gradients have been observed in species occupying DENs, we predicted that species limited by the network also exhibit a downstream increase in genetic diversity.

Methods

Study system

A relatively natural flow regime persists within the upper Gila River catchment in southwestern New Mexico, USA (Fig. 1), because there are few human-engineered stream barriers and a majority of its headwaters are within Gila and Aldo Leopold National Wilderness areas with minimally impacted land use (Propst *et al.* 2008). We selected 16 sample locations representing >200 rkm (river kilometres) with an elevation gradient

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Table 1 Genetic characteristics and selected life history traits of seven native and two non-native Gila River fishes. Genetic summary statistics characterize n individuals per species sampled in 2010 from N of 16 locations corresponding to Fig. 1. Reported are the mean estimates of gene diversity (H_e), observed heterozygosity (H_o), mean allelic richness (A_R) across sites, global F_{ST} with confidence intervals (CI), Mantel r describing the correlation between stream distance and linearized pairwise F_{ST} and global genetic effective size (N_{eS}) based on the sibship method in COLONY (Wang 2009). Maximum body length (L_{Max}) and habitat preference from Olden *et al.* (2006). Genetic summary statistics of *Gila nigra*, *Meda fulgida* and *Tiaroga cobitis* previously reported in Pilger *et al.* (2015)

Species	n (N)	H_e	H_o	A_R	F_{ST} (95% CI)	Mantel r	N_{eS} (95% CI)	L_{Max} (mm)	Habitat
Agosia chrysogaster	397 (13)	0.759	0.746	9.6	0.016 (0.012–0.020)	0.18	332 (275–383)	100	Generalist
Gila nigra	80 (5)	0.627	0.616	5.5	0.027 (0.014–0.041)	−0.24	67 (47–97)	405	Specialist pool
Meda fulgida	170 (6)	0.804	0.734	9.3	0.021 (0.015–0.026)	0.88	167 (128–220)	91	Specialist run/shoal
Rhinichthys osculus	227 (8)	0.705	0.661	8.2	0.042 (0.027–0.062)	0.97*	183 (144–227)	110	Specialist riffle
Tiaroga cobitis	112 (4)	0.730	0.677	10.5	0.008 (0.005–0.011)	0.98*	156 (116–211)	65	Specialist riffle
Catostomus insignis	292 (12)	0.816	0.825	10.5	0.013 (0.009–0.016)	0.59*	285 (239–346)	800	Generalist
Pantosteus clarkii	344 (12)	0.770	0.758	8.5	0.014 (0.011–0.019)	−0.21	190 (152–234)	330	Specialist riffle
Ameiurus natalis†	122 (6)	0.309	0.333	2.2	0.007 (−0.004–0.024)	0.66*	31 (21–52)	470	Generalist
Micropterus dolomieu†	195 (10)	0.469	0.450	2.9	0.036 (0.024–0.048)	0.30	46 (31–68)	690	Specialist pool

*Significant at $P < 0.05$.
†Non-native species.

of nearly 900 m (1161–2059 m above sea level; Table S1, Supporting information). The course of the upper Gila River within the study area is interspersed with valley and canyon-bound reaches with sparse human settlement and agriculture in lower-elevation valley reaches (Whitney *et al.* 2014). The most downstream site was located just below a permanent diversion structure (height ~3 m) that is passable during elevated flows. Several permanent tributaries to the Gila River experience varying levels of intermittency at their confluences with the mainstem; of these, we sampled fish from Blue Creek, Black Canyon, Sapillo Creek and Little Creek (Fig. 1). All tributaries are high-gradient streams surrounded by mountainous landscapes. Downstream of our lowest sample location, the Gila River frequently dries to the degree that immigration of individuals from downstream populations is infrequent during evolutionary time, as supported by phylogeographic studies of three community members (Tibbets & Dowling 1996). Hence, we considered populations of each species in the upper Gila River in New Mexico to be closed to migration from the lower Gila River in Arizona.

Sampling

We sampled each location in October and November of 2010 using a combination of backpack electrofishing and seining (4.6 × 1.2 m, 3.2 mm mesh). During each sampling event, individuals of similar size, typically age-0, for each species were collected in an attempt to include only individuals of the same cohort. Tissue samples for DNA extraction were obtained nonlethally by clipping a small portion (<5 mm^2) of the caudal fin and preserving it in 95% ethanol. Sampled individuals were returned to the stream at the capture site.

Microsatellite genotyping

Genomic DNA was extracted from air-dried fin clips using standard proteinase-K digestion and standard phenol–chloroform extraction (Hillis *et al.* 1996). Microsatellite loci for each species (10 loci for each species except *Catostomus insignis* and *Ameiurus natalis* for which only nine loci were used) were selected and screened for amplification from published sources on focal species or closely related taxa (Table S2, Supporting information). For each species, microsatellite DNA fragments were amplified in 10-μL reactions mixture containing 1.0 μL template DNA (10–50 ng), 1× Colorless GoTaq® Flexi Buffer, 2.5 mM $MgCl_2$ solution, 125 μM dinucleotide triphosphates (dNTPs), 0.4 μM of forward (labelled) and reverse primers and 0.375 units of GoTaq® DNA Polymerase. Multiplex reactions containing up to three loci were optimized depending on annealing temperature, size range and fluorescent label for rapid genotyping of individuals. PCR profiles consisted of 90 °C for 3-min initial denaturation followed by 30 cycles of 90 °C for 30 s, primer-specific annealing temperature (Table S2, Supporting information) for 30 s and 72 °C for 45 s and ending with a final extension at 72 °C for 15 min. Sample preparation for capillary electrophoresis consisted of 1 μL of PCR product mixed with 10 μL formamide and 0.35 μL of HD400 size standard (ABI), then denatured at 90 °C for 5 min. All samples were run on an automated ABI 3130 DNA sequencer and scored using GENEMAPPER software (ABI).

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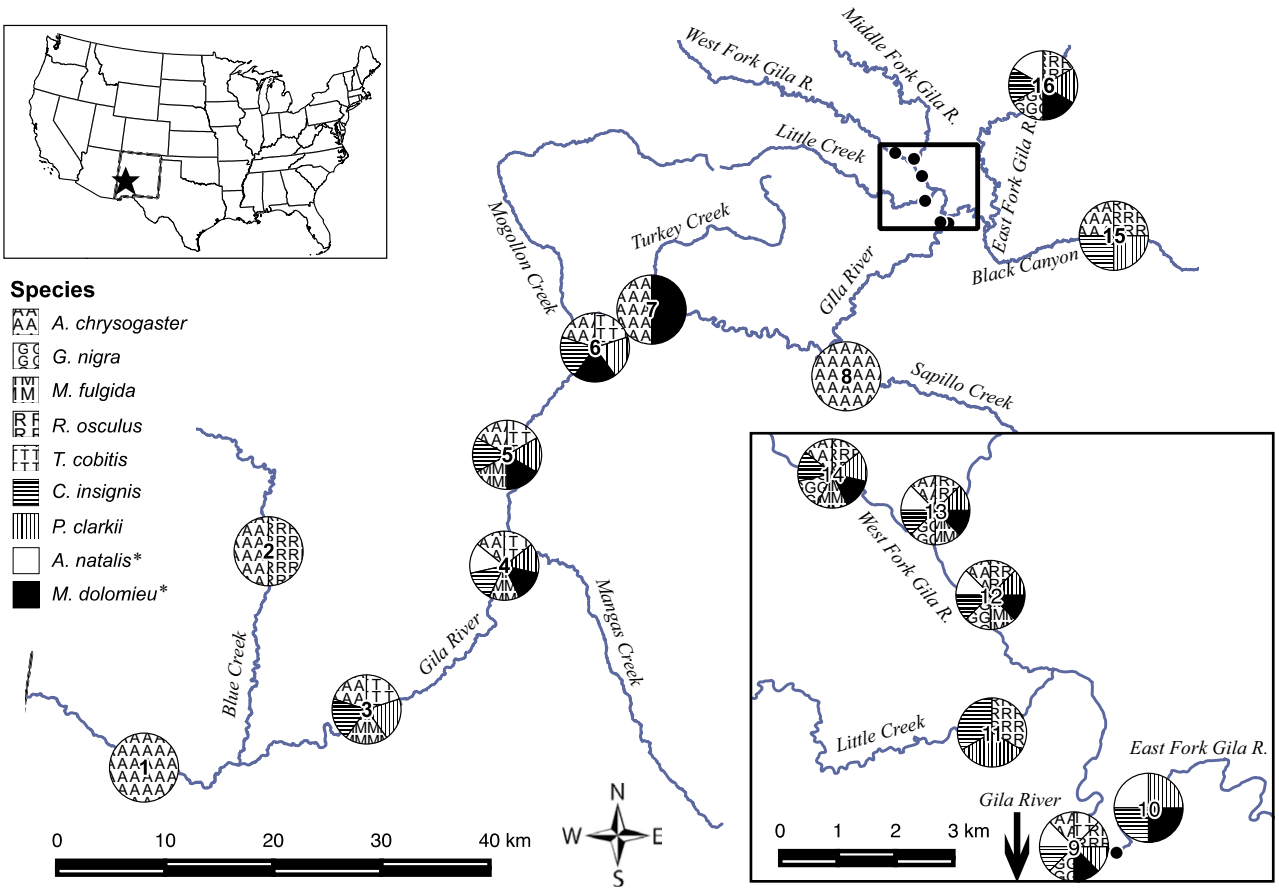

Fig. 1 Sample locations of seven native and two non-native (*) fishes from the upper Gila River catchment, New Mexico, USA. Pie chart slices indicate presence or absence of species at a given site during 2010 sampling. Numbers in the centre of each pie chart correspond to site numbers (Table S1, Supporting information). [Colour figure can be viewed at wileyonlinelibrary.com]

Approximately 10% of samples for each species were reanalysed and scored to validate allele calls and ensure scoring accuracy.

Spatial genetic variation

Tests for departures from Hardy–Weinberg proportions (HWP) by location and a global test for linkage disequilibrium (LD) were performed using GENEPOP (Raymond & Rousset 1995; Rousset 2008). Sequential Bonferroni correction (Rice 1989) was applied to account for inflated type I error rates associated with multiple comparisons. When significant departures from HWP were observed, the program MICROCHECKER (Van Oosterhout *et al.* 2004) was used to examine genotype data for null alleles, scoring errors and large allele dropout which could lead to significant departures from HWP. FSTAT (Goudet 1995) was used to estimate allele frequencies and three diversity statistics, Nei's (1987) unbiased gene diversity (H_E), observed heterozygosity (H_O) and allelic richness (A_r). We evaluated spatial genetic variation for each species using Weir & Cockerham (1984) *F*-statistics. We estimated overall population differentiation, measured as global F_{ST}, for each species using FSTAT, which implements a bootstrap procedure to test the null hypothesis that the observed F_{ST} value is significantly different from zero. We characterized fine-scale genetic structure between samples for each species by estimating F_{ST} values between all pairs of samples using ARLEQUIN 3.0.1 (Excoffier *et al.* 2005). ARLEQUIN implements a permutation test (n = 999 permutations) for significance of pairwise F_{ST} values. We tested each species for conformance to isolation by distance (IBD) by performing Mantel's tests on Slatkin's linearized F_{ST} values ($F_{ST}/(1 - F_{ST})$; Slatkin 1995) and stream distance between sites. Stream distances in kilometres were estimated from Google Earth and $\log_{10}$-transformed to approximate normality for analysis. To assess spatial patterns of genetic diversity and identify genetic diversity gradients, we calculated Pearson's product–moment correlation between allelic richness and distance from the most downstream site (Site #1; Table S1, Supporting

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information). Correlation analyses were performed in R (ver. 2.15.0, R Core Team 2012); Mantel tests were completed using the ECODIST package (Goslee & Urban 2007).

Landscape genetic analyses

We used a landscape genetic approach to evaluate whether or not species exhibited genetic structuring related to network architecture of the Gila River drainage. We hypothesized that riverscape factors such as stream distance, difference in elevation between locations and/or number of intervening confluences between sample locations would have a strong association with linearized F_{ST}. In addition to triangular pairwise F_{ST} and distance ($\log_{10}$-transformed) matrices computed for use in Mantel tests, we populated a third pairwise matrix for each species with the absolute difference in elevation between sample locations. A fourth matrix was populated with number of intervening confluences between pairs of sample locations. Absolute difference in elevation was $\log_{10}$-transformed to more closely approximate normality. We assessed multicollinearity among spatial variables by calculating a variance inflation factor (VIF) for each variable using the R package FMSB (Nakazawa 2015). VIF values >5 suggest multicollinearity among variables (Legendre & Legendre 2012).

The differentiating effect of genetic drift due to low N_e of local populations could increase F_{ST} between populations regardless of landscape factors (Weckworth *et al.* 2013). Pilger *et al.* (2015) identified low N_e (< 350) for *Meda fulgida* and *Gila nigra*, suggesting drift within local populations could have large effects on genetic differentiation among populations for those taxa. Therefore, we quantified variation in N_e among local populations to account for the effects of genetic drift in spatial genetic structure. We estimated effective size for each species at each location using the sibship method (N_{eS}) implemented in COLONY (version 2.0.5.1; Wang 2009). COLONY uses maximum likelihood to estimate probabilities of full- and half-siblings of a sample of individuals taken from a population, from which N_{eS} can be estimated. Pairwise comparisons between local effective sizes (N_{eS}) were quantified by constructing pairwise matrices of the harmonic mean of N_{eS} between sample locations to evaluate the effects of genetic drift on population structuring following Weckworth *et al.* (2013).

We evaluated relationships of extrinsic variables (distance, elevation difference and number of confluences) and differences in genetic effective size (harmonic mean of N_{eS}) on spatial genetic variation using an information theoretic approach in combination with multiple regression on distance matrices (MRM). This approach allowed us to evaluate combinations of explanatory variables while accounting for the nonindependence of distance matrices. Initially, 15 standard linear models (function *lm*) were fit to linearized F_{ST} representing all combinations of explanatory variables for each species except for *Tiaroga cobitis* (Table 2). Only seven models that incorporated distance, elevation and N_{eS} were fit for *T. cobitis*, as this species was only present in mainstem locations of the Gila River and thus was not expected to be influenced by intervening confluences. Akaike's information criteria (AIC) were used to select the best candidate models by simultaneously comparing all models. AIC scores were adjusted for small sample size (AIC_C), and Akaike weights (w_i) were calculated using the AICCMODAVG package in R (Mazerolle 2015). Candidate models with lowest AIC_C ($\Delta AIC_C < 2.0$) and highest weights ($w_i > 0.10$) were retained for interpretation (Burnham & Anderson 2002) and further evaluated using MRM from the ECODIST package (function *MRM*) to identify the contribution of explanatory variables to the overall fit of the model (MRM R^2). Significance of MRM models was assessed by permutation test (1000 permutations).

Results

Total number of individuals assayed per species ranged from 80 *Gila nigra* to 397 *Agosia chrysogaster* (Table 1 and Table S3, Supporting information). *Agosia chrysogaster*, *Pantosteus clarkii* and *Catostomus insignis* were the most widespread, occurring at 13, 12 and 12 of 16 sample locations, respectively (analyses of *C. insignis* were restricted to nine sites where $n > 10$). *Tiaroga cobitis* and *G. nigra* occurred at the fewest sites (four and five sites, respectively). Sample sizes and occurrences from each species were consistent with relative abundance and distribution patterns previously reported for the upper Gila River catchment (Propst *et al.* 2008; Whitney *et al.* 2014, 2016). In all but two species, the number of significant deviations from HWP was consistent with the experiment-wise error rate of $\alpha = 0.05$ (Waples 2015) and application of Bonferroni correction resulted in low frequency of significant comparisons (<0.04). The number of significant HWP tests for *Meda fulgida* and *Tiaroga cobitis* exceeded that expected at $\alpha = 0.05$ and each had 15% of tests significant after correction for multiple comparisons. The majority of significant tests were due to heterozygote deficiency, and MICROCHECKER indicated null alleles as a causal factor. All species had at least one significant test for LD but these were not locus or site specific. Gene diversity, observed heterozygosity and allelic richness were greatest for *C. insignis* (mean $H_E = 0.82$, $H_O = 0.83$, $A_R = 10.5$) and lowest in *A. natalis*

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Table 2 Multiple regression on distance matrices (MRM) for explaining linearized pairwise F_{ST} of Gila River fishes sampled in 2010. Models were ranked using AICc and only models with ΔAICc < 2.0 and AICc w_i > 0.10 are reported, where K is the number of parameters in each model (including intercept and error term). Coefficients are for each explanatory variable included in the model (DIST = stream distance; CONF = number of confluences; N_{eS} = harmonic mean of effective size; ELEV = absolute difference in elevation). MRM R^2 is the amount of variation explained by the model using MRM analysis with P-values for each model based on 1000 permutations

					Coefficients				MRM	
Models	K	ΔAICc	w_i	Cum. w_i	DIST	CONF	N_{eS}	ELEV	R^2	P
Agosia chrysogaster										
CONF + N_{eS} + ELEV	5	0.00	0.24	0.24		6.8E-03	4.0E-04	5.6E-03	0.269	0.099
CONF + ELEV	4	0.24	0.21	0.45		4.8E-03		5.8E-03	0.245	0.042
DIST + CONF	4	0.98	0.15	0.59	5.8E-03	5.4E-03			0.238	0.042
DIST + CONF + N_{eS}	5	1.27	0.13	0.72	5.3E-03	7.3E-03	3.6E-04		0.257	0.124
Meda fulgida										
N_{eS} + ELEV	4	0.00	0.33	0.33			−3.5E-04	2.4E-02	0.781	0.090
DIST + N_{eS}	4	1.15	0.19	0.52	2.3E-02		−4.8E-04		0.764	0.082
ELEV	3	1.47	0.16	0.68				2.3E-02	0.689	0.072
Rhinichthys osculus										
DIST + ELEV	4	0.00	0.29	0.29	1.3E-01			−6.1E-02	0.706	0.002
DIST + CONF + N_{eS}	5	0.58	0.22	0.50	6.9E-02	−2.0E-02	−4.0E-03		0.730	0.010
DIST + N_{eS} + ELEV	5	1.01	0.17	0.68	1.1E-01		−1.9E-03	−4.7E-02	0.726	0.004
DIST + CONF + N_{eS} + ELEV	6	1.77	0.12	0.80	1.0E-01	−1.4E-02	−2.9E-03	−3.3E-02	0.749	0.007
Tiaroga cobitis										
DIST	3	0.00	0.66	0.66	1.4E-02				0.866	0.043
ELEV	3	1.38	0.33	1.00				1.0E-02	0.831	0.046
Gila nigra										
DIST + CONF + N_{eS} + ELEV	6	0.00	1.00	1.00	4.1E-02	−1.5E-02	−5.6E-04	−3.9E-02	0.459	0.920
Catostomus insignis										
N_{eS}	3	0.00	0.29	0.29			−1.1E-03		0.876	0.066
CONF + N_{eS}	4	0.10	0.28	0.57		2.5E-03	−1.1E-03		0.884	0.082
Pantosteus clarkii										
DIST + N_{eS}	4	0.00	0.44	0.44	9.4E-03		−5.2E-04		0.661	0.001
DIST + N_{eS} + ELEV	5	0.70	0.31	0.75	1.4E-02		−5.2E-04	−4.5E-03	0.669	0.001
Ameiurus natalis										
DIST	3	0.00	0.39	0.39	6.1E-03				0.388	0.049
ELEV	3	1.26	0.21	0.60				5.0E-03	0.306	0.111
Micropterus dolomieu										
DIST	3	0.00	0.29	0.29	1.1E-02				0.158	0.159
ELEV	3	0.18	0.27	0.56				9.9E-03	0.148	0.206

(H_E = 0.31, H_O = 0.33, A_R = 2.2; Tables 1 and S3, Supporting information).

All species, except for *Ameiurus natalis*, exhibited significant genetic structuring with global F_{ST} values ranging from 0.008 (*Tiaroga cobitis*) to 0.042 (*Rhinichthys osculus*; Table 1). In testing for IBD, four of nine species had a significant Mantel test for the relationship of stream distance and linearized F_{ST} (Table 1, see Table S4, Supporting information for pairwise F_{ST}s). Four species had significant Pearson's correlations between allelic richness and distance from the most downstream site. Negative significant relationships were observed for *A. chrysogaster* (r = −0.91, t_{11} = −761, P < 0.001) and *M. fulgida* (r = −0.97, t_{11} = −7.81, P = 0.001), indicating higher genetic diversity downstream than upstream. *Rhinichthys osculus* (r = 0.86, t_{11} = 4.07, P = 0.007) and *P. clarkii* (r = 0.60, t_{11} = 2.36, P = 0.040) exhibited the opposite trend. Combined population effective sizes using the sibship method (N_{eS}) ranged from 31 (*A. natalis*) to 332 (*A. chrysogaster*, Table 1).

Landscape genetic analyses

We evaluated the association between extrinsic variables (stream distance, difference in elevation between sites and number of confluences) and genetic drift (harmonic mean of N_{eS} between population pairs) with

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linearized F_{ST} across species. Extrinsic variables did not exhibit strong multicollinearity (range of VIF values = 1.0–3.6) despite stream distance and difference in elevation being correlated (Mantel $r = 0.77$, $P = 0.001$). Model selection with MRM analysis identified five species where extrinsic variables and harmonic mean of N_{eS} accounted for substantial variation ($R^2 \geq 0.66$) in pairwise F_{ST} (Table 2, Fig. 2). Competing models ($\Delta AIC_C < 2.0$ and $w_i > 0.10$) for *M. fulgida*, *R. osculus* and *P. clarkii* identified stream distance and difference in elevation, along with pairwise harmonic mean of N_{eS} for explaining spatial genetic structure of these species. For example, distance and elevation accounted for 71% of *R. osculus* spatial variation but inclusion of confluences or N_{eS} did not appreciably increase R^2. However, inclusion of N_{eS} in competing models for *M. fulgida* and *P. clarkii* did increase explained variation over distance and elevation alone. For *M. fulgida*, the explanatory power of elevation alone ($R^2 = 0.69$) was increased when combined with N_{eS} ($R^2 = 0.78$). Similarly, for *P. clarkii*, distance alone was not a competing model but distance with N_{eS} explained 66% of spatial variation. In *T. cobitis*, distance alone and elevation alone were competing models, each explaining substantial spatial variation (87% and 83%, respectively). Lastly, N_{eS} alone accounted for substantial genetic variation of *C. insignis* ($R^2 = 0.88$), but inclusion of confluences provided only trivial increases in explanatory power.

In four species, predictor variables did not explain a significant amount of variation in pairwise F_{ST} (Table 2). All four competing models for *A. chrysogaster* included number of confluences, which itself accounted for low levels of variation ($R^2 = 0.20$) but was not a competing model. Only one model containing all variables was chosen for best explaining genetic variation in *G. nigra* ($R^2 = 0.46$), but was not significant. Distance and elevation appeared as competing models for both non-native *A. natalis* and *M. dolomieu* but were generally poor predictors of spatial genetic variation. Although distance was the top model for explaining linearized F_{ST} of *A. natalis* ($R^2 = 0.39$), no pairwise F_{ST} values were significant between sites (Table S4, Supporting information). Genetic variation of *M. dolomieu* showed no association with landscape variables or N_{eS}.

Discussion

By adopting a comparative landscape genetic approach that explicitly accounted for the effects of genetic drift among populations on genetic structure, we were able to evaluate the relative roles of extrinsic and intrinsic factors for shaping patterns of neutral genetic variation across nine sympatric fish species. Mantel and MRM analyses indicated stream network architecture predicted spatial genetic connectivity (pairwise F_{ST}) in four Gila River natives, *Meda fulgida*, *Rhinichthys osculus*, *Tiaroga cobitis* and *Pantosteus clarkii*. This is consistent with expectations that gene flow in small-bodied or habitat specialist species (*P. clarkii* is the smaller of the two catostomids) should be constrained by stream network architecture, whereas gene flow in large-bodied or generalist species should not. However, we observed no general trend between body size and global levels of differentiation among species. Additionally, small-bodied *Agosia chrysogaster* and *M. fulgida* exhibited downstream increase in intraspecific genetic diversity, a pattern observed across a diverse array of taxa occupying DENs (Paz-Vinas *et al.* 2015). However, for cases where species occurred at few locations (e.g. *T. cobitis*, *Gila nigra* and *Ameiurus natalis* were present at ≤5 locations), we urge caution when interpreting results due to the low power to detect spatial patterns. For example, the landscape analyses identified stream distance as a predictor for the non-native *A. natalis*, yet global F_{ST} and pairwise F_{ST} values were not significantly different from zero, suggesting that *A. natalis* was essentially panmictic.

Differentiation among populations of species experiencing migration–drift equilibrium (i.e. the rate of gene flow bringing in new alleles matches the rate at which alleles are lost due to genetic drift) is expected to be a function of the distance between them (Hutchison & Templeton 1999). However, genetic drift experienced by a local population can be exacerbated by low N_e, confounding correlations between environmental factors and genetic distance (Weckworth *et al.* 2013). To correct for this, we incorporated pairwise harmonic mean of N_{eS} (calculated for each pair of localities) in our models and found it was present in candidate models for six of nine focal species and was particularly informative for explaining variation in pairwise F_{ST} for *C. insignis* (88%). When combined with distance, harmonic mean of N_{eS} explained substantial variation for *P. clarkii* (73%) and *M. fulgida* (81%). Annual monitoring of the Gila fish assemblage has found abundance of some native species, including *M. fulgida*, to be declining (Propst *et al.* 2008) and, based on this, Pilger *et al.* (2015) suggested genetic drift to be an important force of differentiation for some species. Our finding that local effective size is important for both catostomid species is somewhat surprising given their widespread distributions and greater abundance relative to other focal species. The influence of N_{eS} on pairwise F_{ST} for *C. insignis* and *P. clarkii* can be seen in Fig. 2, whereby the largest pairwise F_{ST} values tended to coincide with much smaller harmonic mean of N_{eS}. These site comparisons were also between mainstem sites and tributaries with small watershed area (e.g. sites 11 and 15 in

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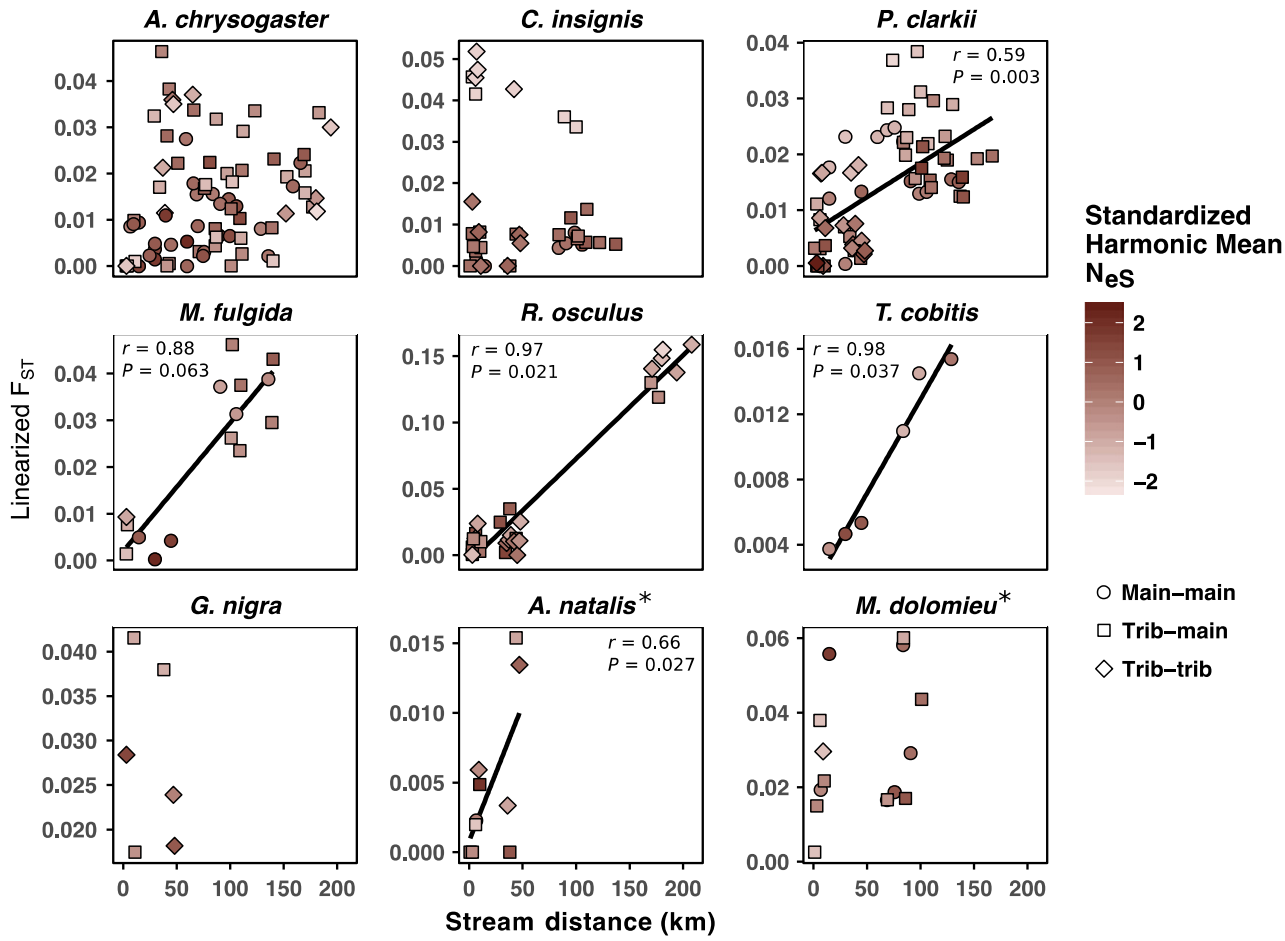

Fig. 2 Scatter plots depicting the relationship between genetic distance (linearized F_{ST}) and linear stream distance (km) for seven native and two non-native (*) Gila River fishes sampled in 2010. Mantel *r* and *P*-values are included for species that exhibited high correlation with stream distance. Symbol fill represents pairwise harmonic mean of N_{eS} between sites; lighter shades indicate smaller harmonic mean N_{eS}, and darker shades indicate larger harmonic mean N_{eS}. Harmonic mean was standardized using z-transformation across species for ease of visual comparison among species. Sample location comparisons (symbol shapes) were classified as mainstem to mainstem (circle), tributary to mainstem (square) or tributary to tributary (diamond) (Table S1, Supporting information). Note the difference in scale on the *y*-axes, and all *x*-axes are identical. [Colour figure can be viewed at wileyonlinelibrary.com]

Fig. 1), indicating these smaller tributaries might support fewer breeding individuals and would thus be subject to stronger drift than larger mainstem populations. Weckworth *et al.* (2013) found that patterns of connectivity of woodland caribou could be influenced by spatial arrangement of preferred habitat and variation in N_e among populations, and accounting for N_e increased correlations with environmental variables. Similarly, in the present study, the effect of stream distance on pairwise F_{ST} of *P. clarkii* was masked until accounting for N_e among sites.

After accounting for extrinsic factors and genetic drift, *A. chrysogaster*, *G. nigra*, *A. natalis* and *M. dolomieu* had substantial unexplained variation in pairwise F_{ST}, suggesting network geometry plays a reduced role in shaping spatial genetic variation for these species. Nonequilibrium population processes such as extirpation and recolonization dynamics (Fields & Taylor 2014) or ecological disturbances (Banks *et al.* 2013) can influence patterns of genetic variation and might lead to genetic differentiation patterns unrelated to network architecture. Extrinsic factors and harmonic mean of N_{eS} were generally weak predictors of *A. chrysogaster* and *G. nigra* differentiation. Temporal instability of genetic structure was observed for *G. nigra* over consecutive years (Pilger *et al.* 2015), and this species was also reported to have high metapopulation extinction probabilities relative to other Gila River species (Pilger 2015; Whitney *et al.* 2016). Alternatively, *A. chrysogaster* is a small-bodied, numerically dominant and widespread species. This species exhibits large demographic fluctuations owing to its short lifespan and 'boom–bust' life

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cycle, presumably an adaptation to highly variable desert streams, making it an excellent colonizer following disturbances (Minckley & Deacon 1968; Minckley & Barber 1971; Whitney *et al.* 2016). Disturbance history and time since colonization (i.e. subpopulation age) are known to obscure genetic patterns and influence distribution of genetic diversity (Haag *et al.* 2005; Cosentino *et al.* 2012). Further characterization of nonequilibrium dynamics of Gila River fishes could shed light on mechanisms influencing genetic differentiation (Pilger 2015).

Although intraspecific spatial genetic variation within both non-native species was poorly explained by extrinsic properties or N_{eS}, *M. dolomieu* and *A. natalis* exhibited contrasting levels of genetic structure. For example, *M. dolomieu*, a habitat specialist, had significant pairwise F_{ST} values from the same localities where the generalist *A. natalis* was collected (site 13 compared with sites 10 and 9), which, as described above, was essentially panmictic. Differences in time since introduction and stocking histories (i.e. the number of distinct founder events) could explain the different genetic patterns between non-natives. Stocking of both species began in the early 1900s and occurred haphazardly until the 1960s and 1970s (records retrieved from New Mexico Department of Game and Fish); however, records were insufficiently detailed to evaluate differences between species. We are unaware of a published account of spatial genetic variation of *A. natalis* or any congener within native or introduced ranges; however, *M. dolomieu* pairwise F_{ST} values in the current study were similar to those reported over a similarly fine spatial extent in the Laurentian Great Lakes region where the species is native (Stepien *et al.* 2007). Stepien *et al.* (2007) observed substantial structure for *M. dolomieu* at fine spatial scales without evidence of IBD. They attributed the high degree of differentiation to life history traits such as high parental investment (nest building and guarding) and high spawning site fidelity. Five microsatellite loci used in the present study were also used by Stepien *et al.* (2007), which allowed direct comparisons across studies. For these five loci, we observed only slightly fewer alleles per locus (1–3 fewer) in the Gila River compared to the entire Laurentian Great Lakes region. In addition, observed heterozygosity was similar across studies (range 0.32–0.56 present study, 0.27–0.59 in Stepien *et al.* (2007)) despite our smaller sample sizes.

Conclusion

Understanding the role that ecological network geometry plays in shaping ecological and evolutionary processes is a rapidly expanding field with many more theoretical and simulation studies than empirical examples. Simulation studies comparing different network geometries, including DENs, provide evidence that the complex spatial arrangements of habitats can increase genetic differentiation among populations and can result in longitudinal genetic diversity gradients (Morrissey & de Kerckhove 2009; Paz-Vinas & Blanchet 2015). By comparing spatial distribution of genetic variation for nine species with differing dispersal abilities and life history traits occupying the same riverscape, we demonstrated that the upper Gila River catchment provides a template which constrains genetic structure for some, but not all, focal species. Native species whose genetic variation is constrained by the network also exhibited high (*Rhinichthys osculus*) and low (*Tiaroga cobitis*) levels of structure that coincided with species distributions in the network; *R. osculus* is restricted to headwaters and tributaries, whereas *T. cobitis* primarily occurred in the Gila mainstem. Furthermore, downstream increases in genetic diversity were observed for *Meda fulgida*, whose structure was constrained by the network, and *Agosia chrysogaster*, whose structure was not associated with the network.

Contrasting spatial patterns of diversity among species suggest that while extrinsic factors may influence the distribution of genetic diversity, intrinsic factors such as local abundance patterns or dispersal propensity may be more relevant to degree of structuring. Gomez-Uchida *et al.* (2009) found network properties such as distance and waterfall barriers to predict genetic patterns for three sympatric salmonids, yet the degree of genetic structuring between species was influenced by intrinsic dispersal abilities and population size. Similarly, genetic patterns related to species dispersal abilities have also been observed for aquatic macroinvertebrates (Phillipsen *et al.* 2015) and amphibians (Mims *et al.* 2015). Our study highlights the fact that there are multiple mechanisms that can influence species' genetic patterns. Thus, predictive models that include extrinsic factors along with intrinsic traits are likely to increase our understanding of how genetic diversity is superimposed on a common landscape.

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Data accessibility

Microsatellite genotypes were deposited to the Dryad Digital Repository: https://doi.org/10.5061/dryad.d5j4m.

All authors contributed to the study design, sample collection and writing the manuscript. T.J.P. performed data collection and analyses and prepared figures.

Supporting information

Additional supporting information may be found in the online version of this article.

Table S1 Latitude, longitude, and elevation of upper Gila River catchment locations sampled during October and November 2010.

Table S2 Summary of genetic screening and cross-species amplification of microsatellite loci for Gila River imperiled fishes collected in 2009 and 2010.

Table S3 Genetic summary statistics for warm-water fishes of the upper Gila River sampled in October and November 2010.

Table S4 Slatkin's linearized F_{ST} values (lower diagonal) for warm-water fishes of the upper Gila River sampled in October and November 2010.

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